MIL-O-13417B
12 April 1983
SUPERSEDING
MIL-O-13417A
24 February 1975

MILITARY SPECIFICATION

OILERS, AIR LINE, FOR PNEUMATIC TOOLS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of oilers for use in the air lines of pneumatic tools.

1.2 Classification. The oilers shall be the following sizes, as specified (see 6.2).

Size 1 - 1/2 pint.
Size 2 - 1 pint.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4930

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking and Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The air-line oilers shall atomize and feed, in any position, any lubricant or liquid grease which is suitable for use in pneumatic tools in the temperature range from plus 110° F to minus 25° F. The oilers shall be equipped with an adjustable oil-metering valve and an oil-filler plug.

3.2 Material. Material shall be of the contractor's current commercial product and shall be subject to all provisions of this specification (see 6.5).

3.2.1 Material deterioration and control. The oilers shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-T-704 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.2.2 Inserts. When the body of the oiler is aluminum, threaded steel inserts shall be used for the oil-filler and oil-regulator access holes and for the air-line connections.

3.3 Temperature conditions. The oilers shall perform as specified herein in any ambient temperature from plus 100° F to minus 25° F when filled with pneumatic-tool lubricant of viscosity suitable for the particular operating temperature.

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3.4 Oil-metering valve. The oil-metering valve shall be removable for cleaning and repairing and shall permit adjustment of the lubricant flow without taking the pressure off the air chamber or interrupting the operation of the pneumatic tool.

3.4.1 Locking device. The lubricator shall be supplied with a locking or tamper-proof, adjusting knob, which shall prevent oil flow changes resulting from shock or vibration.

3.5 Oil-filler plug. An oil-filler plug shall be provided with a hole for bleeding-off pressure inside the oil chamber before threads are disengaged.

3.6 Hydrostatic pressure. When tested as specified in 4.4.2.2, the oilers shall withstand a hydrostatic test pressure of 150 pounds per square inch (psi).

3.7 Performance requirements.

3.7.1 Pressure drop. The pressure drop across the oiler shall be not more than 5 psi (see 4.4.2.3).

3.7.2 Pumping. There shall be no slugs nor an excessive amount of lubricant discharged beyond the normal atomized flow each time the valve is opened (see 4.4.2.4).

3.7.3 Flow. Sufficient lubricant shall pass through the drill to coat the surface of the glass plate within the effective area of the airstream (see 4.4.2.5). There shall be no appreciable difference in the quantity of lubricant deposited on the glass plate because of changing the position of the oiler.

3.7.4 Lubricant leakage. The inside surfaces of the air passage through the oiler shall be dry and free from lubricant (see 4.4.2.6).

3.8 Markings. The oilers shall be identified in accordance with MIL-STD-130 using only the following information, abbreviated wherever possible:

a. Nomenclature

b. Stock number

c. Manufacturer's identification

d. Lubricant capacity

e. Part number

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In addition, the direction of airflow shall be permanently marked on the body of the oiler.

3.9 Treatment and painting. Unless otherwise specified herein, the oilers shall have the contractor's standard commercial finish, which shall be a smooth, even surface, free from runs, sags or foreign material. When specified (see 6.2), the oilers shall be cleaned, treated and painted in accordance with MIL-T-704, type A, color of the finish coat as specified in MIL-T-704.

3.10 Size 1 (1/2-pint capacity). The oiler shall provide constant and ample lubrication to the working parts of pneumatic tools with an air supply as low as 10 cubic feet per minute (cfm). The inlet and outlet shall be threaded with 1/2-inch internal NPT threads. When specified (see 6.2), the oiler shall have universal, quick-disconnect-type hose couplings for both the inlet and outlet connections.

3.11 Size 2 (1-pint capacity). The oiler shall provide constant and ample lubrication to the working parts of pneumatic tools with an air supply as low as 40 cfm. Three-quarter-inch internal NPT threads shall be provided for both the inlet and outlet connections. When specified (see 6.2), the oiler shall have universal, quick-disconnect-type hose couplings for both the inlet and outlet connections.

3.11.1 Hose assembly. When specified (see 6.2), the oiler shall be provided with 10 feet of 3/4-inch air hose, and universal, quick-disconnect-type hose couplings for both ends.

3.12 Workmanship. The quality of workmanship shall be such as to produce oilers that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the oiler.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.1.1 Inspection lot. For the purpose of sampling, a lot shall consist of all oilers of the same size offered for delivery at one time.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The oilers shall be examined as specified herein for the following defects:

Major

101. Materials not as specified, or not corrosion resistant or treated to be made corrosion resistant.
102. Dissimilar metals as defined in MIL-T-704 in intimate contact with each other.
103. Size not as specified.

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104. Threads not as specified.
105. Steel inserts not as specified.
106. Oil-metering valve missing or not as specified.
107. Locking device missing or not as specified.
108. Oil-filler plug missing or not as specified.
109. Quick-disconnect couplings missing, when specified.
110. Hose assembly missing or not as specified.
111. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete or not legible.
202. Treatment and painting not as specified.
203. Material and materials finish or treatment information not available.

4.4.2 Tests.

4.4.2.1 Air leakage. The oilers shall be connected to a 150-psi air line with the discharge end of the oiler plugged. There shall be no indication of air bubbles when the oiler is submerged in a water bath. Evidence of leakage shall constitute failure of this test.

4.4.2.2 Hydrostatic pressure. The discharge end of the oiler shall be plugged and the chamber filled with oil. The metering valve in the oiler shall be open during the test in order to let the pressure build up inside the oil chamber. The inlet end shall be attached to a hydraulic system and the pressure brought up to a pressure of 150 psi. There shall be no signs of leakage in the body of the oiler. Nonconformance to 3.6 shall constitute failure of this test.

4.4.2.3 Pressure drop. The following accessories shall be furnished for this test:

a. Forty-to-100-psi air-pressure regulator with one-inch pipe connections and equipped with air-pressure gage.

b. Direct-reading air meter with range of 30 to 200 cfm.

c. Manometer with copper tubes and fittings for connection to oiler.

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d. Three-quarter-inch throttle valve.

e. One-inch throttle valve.

The air-pressure regulator shall be connected to the air line by joining it to the one-inch throttle valve; in turn, the air meter shall be connected to the pressure regulator. The oiler shall then be connected to the air meter with a tee at each end of the oiler for attaching the manometer tubing. The 3/4-inch throttle valve shall be connected to the discharge tee. With the manometer in position and all joints tight, the air regulator shall be set at 90 psi and the throttle valve adjusted to the proper cfm (50 cfm for the size 1 oiler and 100 cfm for the size 2 oiler) on the air meter scale. The pressure drop on the manometer scale shall be recorded. Nonconformance to 3.7.1 shall constitute failure of this test.

4.4.2.4 Pumping. The oiler shall be attached to an air-supply line 70 pounds per square inch gage (psig) in a horizontal position. A throttle valve with a nozzle capable of passing the required cfm (50 cfm for size 1 oiler and 150 cfm for size 2 oiler) shall be screwed into the discharge end of the oiler. The oiler shall be filled approximately two-thirds full with suitable lubricant and shall be adjusted for normal operation. With the air supply on, the throttle valve shall be opened wide for 5 seconds, then closed for 10 seconds, and then reopened wide for 5 seconds. The cycle shall be repeated for 30 minutes and the oiler checked. Nonconformance to 3.7.2 shall constitute failure of this test.

4.4.2.5 Flow. The oiler shall be filled one-half full with suitable lubricant and tested in the following 4 positions: horizontal with the filler plug in both up and down positions, and with the inlet end up and down 45 degrees from the horizontal. In each of the above positions the oiler shall be used with an appropriate drill under drilling conditions for 20 minutes with an air pressure of 70 psig and an air supply of 55 cfm for the size 1 oiler and 133 cfm for the size 2 oiler. The drill and whip hose shall be cleaned of all excess oil between each test and sufficient oil added to keep the oiler one-half full with lubricant. The exhaust air from the drill shall be checked with a glass plate. Nonconformance to 3.7.3 shall constitute failure of this test.

4.4.2.6 Static pressure test. The oiler shall be completely filled with suitable lubricant and the metering valve set for normal operation. The oiler shall be tested as specified in a. and b. below and the interior air passage checked with a finger or cotton swab. Nonconformance to 3.7.4 shall constitute failure of these tests.

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a. The oiler with discharge end plugged shall be attached to an air line or any other source of supply and allowed to stand for 12 hours under pressure at 150 psi. No air shall leak past the plug in the discharge end.

b. The oiler shall be equipped with a throttle valve on the discharge end. With the throttle valve closed, the oiler shall be charged with air at 150 psig and then the air line shut off. The throttle valve shall then be opened and the oiler allowed to stand for 12 hours.

4.5 Inspection of packaging. Preservation, packing and marking for level A or B shall be inspected in accordance with the Quality Assurance Provisions of PPP-P-40. Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each oiler shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.1.2 Commercial. Each oiler shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Oilers of like description, preserved as specified in 5.1, shall be packed together in accordance with the level A packing requirements of PPP-P-40.

5.2.2 Level B. Oilers of like description, preserved as specified in 5.1, shall be packed together in accordance with the level B packing requirements of PPP-P-40 except fiberboard boxes shall be weather resistant.

5.2.3 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The oilers are intended to be used to keep the moving parts of pneumatic tools constantly lubricated, thus reducing frictional losses and wear.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Size oiler required (see 1.2).

c. When cleaning, treating and painting in accordance with MIL-T-704 is required (see 3.9).

d. Outlet and inlet connections required for size 1 oiler (see 3.10).

e. Outlet and inlet connections required for size 2 oiler (see 3.11).

f. When hose is required for size 2 oiler (see 3.11.1).

g. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the DAR, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of oilers.

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6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army - ME

Review activity:
DLA - CS

User activity:
Navy - MC

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-13417B	Oilers, Air Line, For Pneumatic Tools

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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